



Amplicon Cleaning with ExoSap IT

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Helpful Hints or Important Considerations:

- ExoSap-IT is comprised of Exonuclease I and Shrimp Alkaline Phosphatase in buffer. These two hydrolytic enzymes remove unconsumed dNTPs, primers, and other single-stranded DNA that remain in the PCR product mixture that would otherwise interfere with sequencing.
- There are two types of ExoSap-IT, Regular and Express. Both protocols are below. Know which type you are using before using a thermocycler.
- ExoSap-IT Express can be obtained from the reagent freezer in NHB W133; as of 2022 LAB no longer stocks ExoSap-IT Regular. Please sign out all reagents you take using an LAB iPad (one located on the wall to the left of the reagent freezer in NHB W133) with your 4-digit LAB-assigned PIN.

Materials & Reagents:

- Nuclease Free Water
- PCR Product (that has been checked for amplification success using gel electrophoresis)
- ExoSap-IT Enzyme Mix or ExoSap-IT Express Enzyme Mix (Affymetrix)

Protocol:

1. Determine how many amplicons to clean, and include ~ 5% for pipetting loss, when calculating your cocktail. For example, to clean all wells in a 96-well plate, make the volume required for 100 samples (96 plus 4 for pipetting loss).
2. In a 0.5 ml microcentrifuge tube, mix 0.4 μ l ExoSap-IT and 1.6 μ l nuclease-free water per sample (results in a 1:5 dilution). For example, if you are purifying a 96-well plate of PCR products, then you would mix 40 μ l ExoSAP-IT to 160 μ l nuclease-free water. Note¹: ExoSap-IT is suspended in glycerol, making it particularly viscous. Pipette carefully and slowly to transfer it. Note²: We dilute ExoSap-IT for use compared with the manufacturer's suggestion as we have found this to be as effective.
3. Add 2 μ l ExoSap-IT cocktail to each PCR product. Spin down plate.
4. Incubate plate on thermocycler.
 - a. Regular ExoSAP-IT: Incubate plate at 37°C for 30 minutes, followed by 80°C for 20 minutes to deactivate the enzyme. This product is ready for sequencing.
 - b. Express ExoSAP-IT: Incubate plate at 37°C for 4 minutes, followed by 80°C for 1 minute to deactivate the enzyme. This product is ready for sequencing.
5. Store for < 1 week at 4°C, or -20°C for longer than one week.